A Si-synergic hydrovoltaic electricity generator derived from biogenetic carbonized rice husk to optimize hydrovoltaic current

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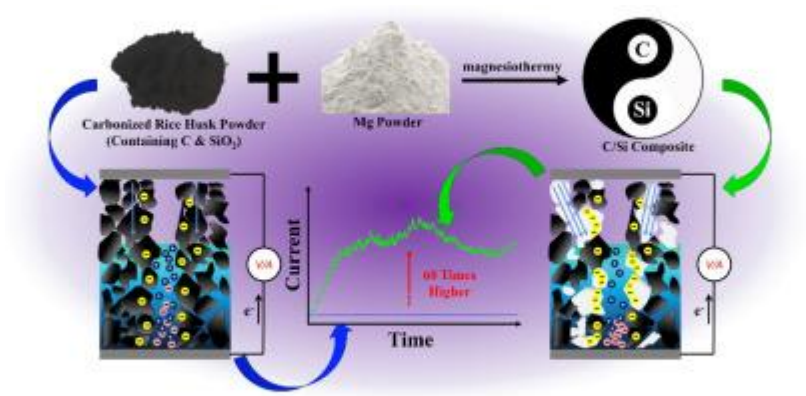
Abstract

In recent years, the research on hydrovoltaic conversion has been rapidly progressed, showing potential prospects in the field of power generation. Despite the encouraging strides, numerous formidable challenges still persist. The foremost impediment lies in the paucity of techniques is to enhance the induced current which generally determines the output power of hydrovoltaic generators. In our previous work, the carbonized rice husk (CRH), a natural biologic material composed of C and SiO_2, has been proved as a promising hydrovoltaic agent for its sensitive response on moisture, but also suffers the same defect with negligible current at only nanoamps level. In this case, this work has provided a progressive strategy on optimizing hydrovoltaic current on the basis of carbonized rice husk device by controlling its composition and structure. The magnesiothermy was employed to partially reduce the amorphous SiO_2 of carbonized rice husks to get a CRH/Si composite. The as-formed Si structures that continuously dispersed across the hydrovoltaic system effectively lower the internal resistance of device and optimize the process of charge migration. As a result, the as-derived CRH/Si device has acquired a sustained hydrovoltaic current up to 1.2 μA, which reveals 60 times increase than that of pre-treatment state. This work also offers a referable route that to enhance induced current in other hydrovoltaic system with poor conductivity by introducing semi-conductive Si nanostructure.

Graphical abstract



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Introduction

With the exacerbation of energy crisis [1] and environmental issues [2], particularly the climate change [3], traditional fossil fuels have become inadequate to meet the developmental demands of contemporary society. Consequently, new energy systems, such as, solar energy [4,5], wind energy [6] and hydrogen energy [7,8], have emerged as the promising techniques for the past few years. Water, being the most original and renewable energy supporter, also possesses immense potential with its unmatched reserves of kinetic energy and internal energy. Currently, river energy [9] and ocean energy [10] are the primary methods employed for harnessing water energy via the conversion of mechanical energy to electricity, but always suffer relatively low conversion efficiency and cause damage to the environment. In recent years, nano-scale hydrovoltaic electricity generators (HEGs) have attracted great attention due to their competitive efficiency, extensive application and simplicity in operation [11]. Among them, the form of power generation from natural evaporation of water has gained more research interest ascribed to the features of straightforward operational procedures and low environmental requirement. The mechanism behind evaporative power generation lies in the combined effects of evaporation process and streaming potential [12]. When water flows through the nanochannels, influenced by the Zeta potential of the material surface, an electric double layer (EDL) is formed consisting of a Stern layer opposing the surface potential and a diffusion layer with the same surface potential [13]. If the channel size is smaller than the Debye length, there would be an EDL overlap resulting in rejection of similar charge, while only carriers with opposite charge can pass through the channel, leading to generation of streaming potential [12]. During the evaporation process, water takes charge across the channels, leading to the generation of electrons or holes within the material and subsequently establishing an evaporation driven potential [12].

Since carbon nanotubes were firstly reported as a potential hydrovoltaic materials in 2003 [14], the field has witnessed a rapid evolution, encompassing a comprehensive material system with graphene [15], nano carbon black [16] and other carbon-based materials [17,18]. These carbon family materials are typically received more attention with their large specific surface area, abundant delocalized electrons and typical EDL effect. Afterwards, a large count of new systems also has been employed to fabricate HEGs, including metal oxides such as Al_2O_3 [19] and TiO_2 [20], semiconductor nanomaterials like Si nanowires [21], SiC [22] and MoS_2 [23], polymers like polypyrrole nanowires [24] and even some biomass materials like protein [25] and bacteria [26]. Benefit from the design of microstructure and construction of nanochannels, these as-reported materials generally show mentionable output voltage range from millivolt to volt level, but most of them always suffer from low induced current with nanoamps (nA). In brief, it is meaningful to explore strategies focus on the enhancement of hydrovoltaic current on the basis of existing system, especially for the most widely carbon materials.

Carbonized rice husk (CRH), a natural biomass carbon which contains amorphous C and SiO_2, has the benefits of easy preparation and low costs [27,28]. It has been proved to be a potential hydrovoltaic material in our previous work [29]. It reveals featured hydrophilicity, steady output voltage and remarkable response on moisture variation for breath monitoring, but also demonstrates the typical challenge with negligible current ascribed to the extreme resistance of the existent SiO_2 component. In contrast with the dielectric SiO_2, silicon containing materials (such as Si, Si/C composites, etc.) are extensively applicated in energy [30], electronics [31], chemical [32] and other fields [33,34]. It is worth mentioning that silicon-based materials are gradually perceived as the most potential negative material in the lithium-ion battery [35,36], which is generally related to the hydrovoltaic effect. The suitable carrier mobility of Si logically endows it a feasible way on the optimization of induced current during hydrovoltaic conversion. Furthermore, in consideration of the abundant amount of silica in carbonized rice husk, the route that in-situ preparing Si/C composites from CRH deservedly becomes clear in the face of low hydrovoltaic current, and magnesiothermy has been proved to be one of the ideal solutions for the preparation of Si/C composites from SiO_2 reduction [28,37].

Herein, our efforts in this work are mainly dedicated to optimize hydrovoltaic current of CRH HEG by adjusting its component. The typical magnesiothermy was taken to in-situ prepare nano-structured Si in CRH with partial reduction of SiO_2, which consequently acquires a CRH/Si composite. The in-situ process makes Si nanostructure uniformly dispersed in the CRH body, and effectively lowers the internal resistance of CRH/Si composite. The partial existent of SiO_2 also enables this composite to get good hydrophilicity similar with the original CRH. Ascribed to the as-mentioned features, the

CRH/Si HEG has exhibited a hydrovoltaic current of about 60 times higher than that one without magnesiothermy treatment. Furthermore, from the COMSOL simulation of space charge distribution in C/Si composite model, it also shows that introducing Si nanostructures in carbon material can effectively accumulate charges around Si, so that to adjust the process of carrier migration and enhance hydrovoltaic current.

Experimental materials and reagents

Rice husk (Hubei, China), terpineol (analytical pure, Shanghai, China), ethyl cellulose (analytical pure, Shanghai, China), glass microfiber filter paper (Whatman GF/C), conductive copper wire (Guangdong, China), epoxy resin adhesive (Kunshan Green Circulation Electronic Materials Co., Ltd, Jiangsu, China), magnesium powder (Shanghai, China). All chemicals were used without further purification.

Preparation of carbonized rice husk

Rice husks were thoroughly washed and subsequently dried. Then, the dried husks were placed into a

Results and discussions

The concept of preparing carbon-silicon composite reinforced CRH HEG is shown in Fig. 2a. By treating with magnesiothermy with the different CRH/Mg ratio, carbon-silicon composite based on CRH was fabricated. Fig. 2b shows the XRD pattern of pure CRH and CRH/Si powder. It reveals an obvious variation on phase composition after magnesium thermal treatment. The feeble peaks at ~22°, ~44°, and broad peak range from 15° to 30° in all samples represents the characteristic diffraction of amorphous SiO

Conclusions

In this work, the induced current of carbonized rice husk (CRH) hydrovoltaic device was successfully enhanced by controlling the composition and structure. The dielectric SiO_2 in CRH was partially reduced to nano-structured silicon by employing magnesiothermy to prepare CRH/Si composites. The addition of Mg during magnesium thermal process directly affects the formation and distribution of silicon structure. When introducing the ratio of Mg to 0.4, the composites possess a featured structure

CRediT authorship contribution statement

Wei Fang: Writing – review & editing, Supervision, Methodology. **Haoyu Zhang:** Writing – original draft, Visualization, Validation. **Shangxi Liu:** Writing – original draft, Visualization, Validation. **Hui Chen:** Formal analysis, Data curation. **Xing Du:** Formal analysis, Data curation. **Xuan He:** Formal analysis, Data curation. **Weixin Li:** Formal analysis, Data curation. **Daheng Wang:** Formal analysis, Data curation. **Lei Zhao:** Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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